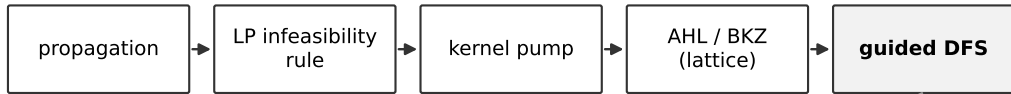
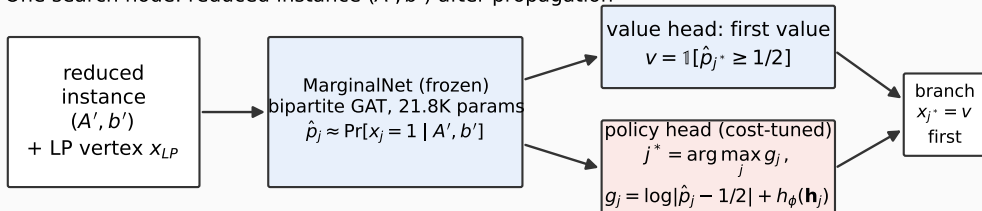


Deployed cascade (sound stages first; the learned components act only in the last stage)



One search node: reduced instance  $(A', b')$  after propagation



Stage 1 (supervised, exact conditional marginals) trains the network and the value rule; Stage 2 (actor-critic on exact subtree costs) trains only  $h_\phi$ . Neither fixes a variable: soundness rests on propagation and LP-probing; a wrong prediction costs backtracking only.